

DERAJKUMAR S

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CAREER OBJECTIVE

Motivated B.E. Computer Science Engineering student with a strong passion for Full Stack Development, Cybersecurity Research, and AI-driven solutions. Enthusiastic about building end-to-end web applications, conducting security vulnerability assessments, and developing intelligent AI systems that drive real-world impact. Equipped with relevant certifications in cybersecurity and hands-on AI project experience, eager to contribute to innovative and security-conscious environments.

AREAS OF ENTHUSIASM

- Full Stack Developer: Building scalable, responsive web applications using modern frontend and backend technologies.
- Cybersecurity Research Analyst: Identifying vulnerabilities, performing penetration testing, and contributing to network security research.
- AI Developer: Designing and deploying AI/ML models for real-world problem solving across domains including environment, health, and safety.

EXPERIENCE

Project Engineering Intern — SRF Limited

Internship | Jun 2026 – Jul 2026 | Chennai, Tamil Nadu, India | On-site

- Completed a one-month internship as a Project Engineering Intern, gaining hands-on exposure to IT infrastructure management and computer information systems.
- Worked on-site with cross-functional teams, applying core engineering and systems concepts in a live industrial environment.

EDUCATION

B.E. – Computer Science Engineering — MNM Jain Engineering College, Chennai (Anna University)

2023 – Present | Current CGPA: 8.1 / 10

12th Grade — St. Joseph Hr. Sec. School, Chengalpattu

May 2023 | Score: 433 / 600

11th Grade — St. Joseph Hr. Sec. School, Chengalpattu

May 2022 | Score: 434 / 600

TECHNICAL SKILLS

- Programming Languages: Python, JavaScript, C, C++
- Web Development: HTML, CSS, JavaScript, Bash Scripting, Frontend Development
- Cybersecurity: Network Security, Risk Analysis, Vulnerability Assessment, Penetration Testing, CTF, Cyber Forensics
- AI / Machine Learning: Computer Vision, LSTM Forecasting, Satellite Data Analysis, AI-powered Automation
- Tools & Platforms: Git, Linux, Bash, VS Code

PROJECTS

WALL-E & EVE – AI for Plastic Pyrolysis — Chennai

September 2024

- Developed AI-powered bots to optimize plastic pyrolysis into biofuels.
- Reduced reaction variability by 25% through intelligent process control.
- Boosted energy efficiency by 15–18% using predictive AI models.

Quality Control for Food Manufacturing — Chennai

May 2025

- Built a smart AI system to automate food quality inspection workflows.
- Achieved 95% defect detection accuracy with real-time contamination alerts.

AEROVISION – AI Powered Air Quality Monitoring Platform — Chennai

June 2025

- Developed an AI-based platform for PM pollution monitoring using AOD and weather data.

- Achieved high spatial accuracy for pollution mapping across regions.
- Enabled 3-day AQI forecasting using LSTM deep learning models.

FORESTGUARD-X – Forest Fire Detection & Simulation System — Chennai

June 2025

- Built a forest fire detection and spread simulation system using satellite and weather data.
- Predicted fire risk with over 85% accuracy.
- Generated hourly fire spread maps in 30m raster format for emergency response.

StegoChat – Steganographic Web Chat — Chennai

2026 | github.com/Derajkumar/steganography-chat

- Built a web chat that hides encrypted messages inside CSS files using End-of-Line (EOL) steganography combined with SHA-256 for integrity and encryption.
- Engineered a React 19 + Vite frontend with an Express/TypeScript backend and a Python-based steganography and cryptography engine, styled with a dark 'blueberry' chat UI.
- Implemented multi-channel notifications (WhatsApp, SMS, Email) so recipients can be reached without ever exposing the underlying covert channel.

AI-Powered Digital Twins for Omnidirectional Robotic Systems — SRM Institute of Science and Technology (SRMIST-KTR)

September 2026 | 3-Day Hands-on Workshop & Hackathon

- Built a Unity-based digital environment and an omnidirectional robot model, simulating physical robot behavior as a digital twin.
- Trained an image-based command-and-response AI model and implemented it across both the physical robot and its digital twin.
- Completed as part of a 3-day hands-on workshop-hackathon hosted by the Department of Computing Technologies, School of Computing, College of Engineering and Technology, SRMIST-KTR, under the mentorship of Ashokkumar Manisekaran.

ACHIEVEMENTS

- Best Innovation Idea Award – St. Joseph Institute of Technology (Cash Prize: Rs. 2,000)
- 2nd Place – National Conference at Sathyabama Institute of Technology
- Finalist – Ideathon2k25, B.S. Abdur Rahman Crescent Institute of Science and Technology
- Participant – Smart India Hackathon 2024 (SIH2024)
- 1st Place in Video Editing & 3rd Place in Idea Pitching – National Science Day 2024 (Theme: Indigenous Technologies for Viksit Bharat)

CERTIFICATIONS

- Cyber Forensics – Great Learning
- Systems Engineering Culture: Cybersecurity – Skillsoft
- Build Your First Agent with Agent Development Kit (ADK) – Google Skills
- Build Agents with Agent Development Kit (ADK) – Google Skills
- Engineer AI Agents with Agent Development Kit (ADK) – Google Skills

LANGUAGES SPOKEN

English | Tamil | Telugu | Hindi

STRENGTHS

Problem Solving | Fast Learner | Creative Thinking | Cybersecurity Enthusiast | Technical Curiosity | Team Collaboration

INTERESTS

Photography | Creative Writing & Poetry | Cycling | CTF Player